GenVoice Care 2.0: Production-Ready AI-Powered Voice Assistant for Elderly Wellness Monitoring with Real-Time Caregiver Coordination

Akshaya G, Sruthi E, Archana V, Dr. R. Anandan
B.Tech Scholar, Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Vels Institute of Science, Technology and Advanced Studies (VISTAS), Pallavaram, Chennai, Tamil Nadu, India

Abstract :
A large number of people are getting older around the world. In India, the number of older adults is expected to reach 319 million by 2050, making up about 20% of the total population. At the same time, there are many jobs related to elder care that remain unfilled across the country. Traditional care systems often fail to meet all the needs of the elderly at once. They lack effective monitoring and fail to assist in proper family care management. In order to solve these problems, we created GenVoice Care 2.0. We have also put in place. This is a mobile application system designed for real-world use. It offers voice-based AI chat, live wellness assessments, and support for caregiver coordination. The system operates on the Railway cloud platform. It consists of three interconnected components. First, there is an Android app specifically tailored for seniors. It uses natural voice input through Android's SpeechRecognizer API. Second, there is a Backend server that runs on cloud using FastAPI framework with total 14 ready-to-use REST API end-points. Third, there is an Android caregiver's dashboard with color-coding and real time monitoring system. Forth, for voice communication part, we have used the Groq's external GPT-OSS-120B model. During testing, it provided responses within 2 to 3 seconds, enabling a more natural and smooth interaction for older adults. For detecting health issues, we implemented a rule-based keyword check. It matches spoken words against six medical terms: pain, hurt, ache, bad, terrible, and awful. It then assigns a wellness score on an 8-point scale, with 8 for Good, 6 for Fair, and 4 for Poor. The backend showed 99.6% API reliability. During production testing, average response times remained between 600 to 700 milliseconds across all endpoints. We also tested keyword detection. We conducted seven check-in scenarios. Each one resulted in the correct classification. This yielded 100% accuracy. Of these, five scenarios involved health concerns, and they were correctly identified. The other two scenarios did not involve health concerns, and they were also correctly identified. This led to zero false positives and zero false negatives. The caregiver dashboard displays alerts in three priority levels: red for high priority, orange for medium, and green for normal. Data updates after a manual refresh and retrieval take about three seconds. This system is currently in production and is accessible at devoted-miracle-production-23ef.up.railway.app. It is available 24 hours a day, seven days a week. This supports the notion that voice-based monitoring for elderly users can function effectively on a larger scale. The current version is a working prototype in a cloud environment.

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